

Week 3: Seller and Inventory Management System

Scenario:

The e-commerce company sells products supplied by multiple sellers. The system should maintain seller details, track products supplied by each seller, and monitor inventory levels to ensure product availability.

1. Create Seller and Inventory Tables

```
SQL> CREATE TABLE Seller_Details(Seller_Code VARCHAR(7), Seller_Title VARCHAR(25),
  2 Contact_Email VARCHAR(30), Phone_Number number(10), Store_Address VARCHAR(45));

Table created.

SQL> CREATE TABLE Stock_Inventory(Inventory_ID VARCHAR(7),
  2 Product_ID VARCHAR(7), Seller_Code VARCHAR(7),
  3 Stock_Quantity number, Stock_status VARCHAR(11),
  4 Last_Updated DATE);

Table created.
```

2. Establish Relationships

Seller and Product

One seller can supply multiple products.

One product can be supplied by one or multiple sellers.

Relationship:

Seller → Product is one to many

Product and Inventory

Each product has inventory information.

Inventory tracks available stock.

Relationship:

Product → Inventory is one to one

Primary Key added for Both Tables:

```
SQL> ALTER TABLE Seller_Details ADD CONSTRAINT PK_Seller PRIMARY KEY(Seller_Code);

Table altered.

SQL> ALTER TABLE Stock_Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY(Inventory_ID);

Table altered.
```

Foreign Key for Inventory Table:

```
SQL> ALTER TABLE Stock_Inventory ADD
  2 CONSTRAINT FK_Prod FOREIGN KEY(Product_ID) REFERENCES Product_Catalog(Prod_ID);

Table altered.

SQL> ALTER TABLE Stock_Inventory ADD
  2 CONSTRAINT FK_Seller FOREIGN KEY(Seller_Code) REFERENCES Seller_Details(Seller_Code);

Table altered.
```

Constraints to make Unique Email ID:

```
SQL> ALTER TABLE Seller_Details ADD
      2 CONSTRAINT Un_mail UNIQUE(Contact_Email);

Table altered.
```

To make Email NOT NULL

```
SQL> ALTER TABLE Seller_Details MODIFY Contact_Email NOT NULL;

Table altered.
```

Add Constraint to Check the Stock Quantity is Positive

```
SQL> ALTER TABLE Stock_Inventory ADD
      2 CONSTRAINT CK_Quan CHECK(Stock_Quantity >=0);

Table altered.
```

Insert Values to Seller Table

```
SQL> INSERT INTO Seller_Details VALUES('&Seller_Code', '&Seller_Title', '&Contact_Email', &Phone_Number, '&Store_Address')
      2 ;
Enter value for seller_code: VND-701
Enter value for seller_title: Apex Logistics & Supplies
Enter value for contact_email: contact@apexlogistics.in
Enter value for phone_number: 9840112233
Enter value for store_address: 14, Industrial Estate, Guindy, Chennai
old 1: INSERT INTO Seller_Details VALUES('&Seller_Code', '&Seller_Title', '&Contact_Email', &Phone_Number, '&Store_Address')
new 1: INSERT INTO Seller_Details VALUES('VND-701', 'Apex Logistics & Supplies', 'contact@apexlogistics.in', 9840112233, '14, Industrial Estate, Guindy, Chennai')

1 row created.
```

```
SQL> SELECT * FROM Seller_Details;
```

SELLER_COD	SELLER_TITLE	CONTACT_EMAIL	PHONE_NUMBER	STORE_ADDRESS
VND-701	Apex Logistics & Supplies	contact@apexlogistics.in	9840112233	14, Industrial Estate, Guindy, Chennai
VND-702	Horizon Retail Partners	sales@horizonretail.com	9840223344	88, Cross Cut Road, Gandhipuram, Coimbatore
VND-703	Prime Source Global	info@primesource.net	9840334455	45, West Masi Street, Madurai
VND-704	Zenith Warehousing Corp	support@zenithcorp.org	9840445566	102, Salai Road, Woraiyur, Trichy

Insert Values to Inventory Table

```
SQL> INSERT INTO Stock_Inventory VALUES('&Stock_ID', '&Prod_ID', '&Seller_Code',
2 &Stock_Quantity, '&Stock_status', TO_DATE('&Last_upd', 'YYYY-MM-DD'));
Enter value for stock_id: STK-901
Enter value for prod_id: PRD-101
Enter value for seller_code: VND-701
old 1: INSERT INTO Stock_Inventory VALUES('&Stock_ID', '&Prod_ID', '&Seller_Code',
new 1: INSERT INTO Stock_Inventory VALUES('STK-901', 'PRD-101', 'VND-701',
Enter value for stock_quantity: 6
Enter value for stock_status: Available
Enter value for last_upd: 2026-09-18
old 2: &Stock_Quantity, '&Stock_status', TO_DATE('&Last_upd', 'YYYY-MM-DD'))
new 2: 6, 'Available', TO_DATE('2026-09-18', 'YYYY-MM-DD'))

1 row created.
```

```
SQL> SELECT * FROM Stock_Inventory;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
-----
STK-901 PRD-101 VND-701          6 Available  18-SEP-26
STK-902 PRD-101 VND-702          5 Available  19-SEP-26
STK-903 PRD-102 VND-703          0 Unavailable 20-SEP-26
STK-904 PRD-201 VND-704          7 Available  21-SEP-26
```

Report 1: Seller-wise Product Report

Display:

- Seller name
- Number of products supplied
- Available stock

```
SQL> SELECT s.Seller_Title,
2 COUNT(i.Product_ID) AS Number_of_Products,
3 SUM(i.Stock_Quantity) AS Available_Stock
4 FROM Seller_Details s
5 JOIN Stock_Inventory i ON s.Seller_Code = i.Seller_Code
6 WHERE i.Stock_Status = 'Available'
7 GROUP BY s.Seller_Title;

SELLER_TITLE                NUMBER_OF_PRODUCTS AVAILABLE_STOCK
-----
Apex Logistics & Supplies          1              6
Horizon Retail Partners            1              5
Zenith Warehousing Corp            1              7
```

Use Joins to get details from Seller and Inventory Table.

Report 2: Stock Availability Report

Display:

- Product name
- Stock quantity
- Stock status

```
SQL> SELECT p.Prod_Name, i.Stock_Quantity, i.Stock_Status
2 FROM Product_Catalog p
3 JOIN Stock_Inventory i ON p.Prod_ID = i.Product_ID;

PROD_NAME                STOCK_QUANTITY STOCK_STATUS
-----
Ultrabook Pro 15          6 Available
Ultrabook Pro 15          5 Available
Mechanical Keyboard       0 Unavailable
Trail Running Shoes       7 Available
```

Report 3: Inventory Analysis

Find:

- Total products available

```
SQL> SELECT COUNT(*) AS Total_Available_Products
2 FROM Stock_Inventory
3 WHERE Stock_Status = 'Available';

TOTAL_AVAILABLE_PRODUCTS
-----
3
```

- Products out of stock

```
SQL> SELECT COUNT(*) AS OutOfStock_Products
2 FROM Stock_Inventory
3 WHERE Stock_Status = 'Out of Stock';

OUTOFSTOCK_PRODUCTS
-----
0
```

- Highest stocked products

```
SQL> SELECT p.Prod_Name, i.Stock_Quantity
2 FROM Product_Catalog p
3 JOIN Stock_Inventory i ON p.Prod_ID = i.Product_ID
4 WHERE i.Stock_Quantity = (SELECT MAX(Stock_Quantity) FROM Stock_Inventory);

PROD_NAME                STOCK_QUANTITY
-----
Trail Running Shoes       7
```

- Average inventory quantity

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory
2 FROM Stock_Inventory;

AVERAGE_INVENTORY
-----
4.5
```